

SDED v1.2 Revision B: Specification and Approved Libraries

Version: 1.2 Revision B

Author: Brian Wijaya

Date: October 23, 2025, 10:49 AM CST

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DOI: (to be assigned)

Prefatory Note: SDED v1.2 Revision B introduces domain-specific Approved Libraries to standardize terminology across multiple fields by using syntactic replacement of eponymous terms with clear, descriptive equivalents. The goal of this update is to eliminate naming ambiguity and honor the SDED v1.1 principles while expanding coverage. All eponymous terms have been systematically replaced with substantive descriptors that convey meaning, with the original name noted for historical context. Each library is organized by subdomain with thematic keyword groupings for efficient reference. This approach promotes clarity and consistency in technical communication across domains.

Pronunciation: SDED is read "ess-ded," or informally "S Dead," reflecting the protocol's goal of replacing dead-end eponyms with living descriptive clarity.

1. Scope and Proportioning of Coverage (v1.2)

Version 1.2 broadens the scope of the Standard Descriptive Eponym Dictionary (SDED) to encompass a wider range of academic and technical domains. AV-1 spans core STEM fields (Mathematics, Physics, Chemistry, Computer Science, Engineering, Medicine) and select areas of the humanities (Linguistics and Philosophy), reflecting a balanced coverage. Each domain's inclusion in AV-1 is proportioned based on the prevalence of eponymous terminology in that field and the impact on cross-disciplinary communication. Emphasis is placed on high-frequency and foundational terms in each domain to maximize the protocol's utility. While STEM domains constitute the majority of entries (reflecting their heavy use of eponyms in laws, theorems, and units), v1.2 also allocates entries to humanities and social sciences to ensure broad applicability. The scope remains extensible: future revisions may incorporate additional domains or subdomains as needed, maintaining proportional representation to avoid any domain dominance. In summary, SDED v1.2 aims for comprehensive yet balanced coverage, ensuring that no field with significant eponym usage is overlooked, and each included domain is covered in proportion to its historical and pedagogical reliance on eponyms.

2. Syntactic Equivalence Principle (Addendum to v1.1)

SDED v1.2 introduces the **Syntactic Equivalence Principle**, an addendum to the guidelines established in v1.1. This principle asserts that each eponymous term and its approved descriptive replacement are to be treated as syntactically interchangeable in any context. In practical terms, a sentence using an eponym can be rewritten using the approved descriptor without altering the grammatical structure or meaning. The descriptive term stands in for the eponym one-to-one, serving as a direct substitute rather than a mere explanation. This equivalence extends to plurality, possessive forms, and adjectives derived from the term, ensuring consistency in usage. By formalizing eponym-descriptor pairs as syntactic equals, v1.2 guarantees that adopting the approved vocabulary will not disrupt the flow of technical documents or discourse. This addendum complements v1.1's semantic focus (ensuring terms carry equivalent meaning) with a syntactic guarantee, thereby fully preserving both meaning and grammatical function when replacing eponyms with their descriptive alternatives.

3. Structural Rules for Glossary Generation

All entries in the Approved Vocabulary adhere to a strict schema to maximize information density and clarity. The structural rules for generating the glossary entries are as follows:

- **Descriptive Replacement:** For each eponymous term, create a concise descriptive term that captures the essence of the concept. Any informative noun from the original name (e.g., "theorem", "law", "disease", "algorithm") is retained in the new term to indicate the type of concept.
- **Original Name and Date:** After the descriptive term, the original eponym (in full, including any first names or acronyms expanded) is provided in parentheses, suffixed with the trademark symbol TM to denote it as a historical name, followed by \"est.\" and the year of establishment or first publication. This preserves the historical attribution without using the name as the primary identifier.

Using \"Fourier TransformTM\" as a joke or critique of scientific naming conventions is legally safe because trademark law only applies to commercial source identifiers, not to expressive commentary.

Adding \"TM\" in that context is an act of parody, not commerce. Parody is protected because it mimics form to make its point; courts consistently recognize that imitating a mark's appearance is lawful when used for humor, criticism, or commentary and not likely to cause confusion.

By writing \"Fourier TransformTM,\" one is not claiming ownership but highlighting the absurdity of privatizing scientific concepts. The act is expressive, non-commercial, and protected under U.S. free-speech doctrine.

This is not mean-spirited; it parallels the linguistic reclamation of descriptive terms such as \"tissue\" regaining primacy over the brand \"Kleenex.\" Such reclamation returns language to the commons.

- **Library Format:** Each domain's Approved Library uses a compact, reference-style format. Subdomains are introduced with bracketed headers (e.g., [ANALYSIS]) followed by thematic keyword descriptors (e.g., Functions. Series. Continuity.). Term entries are listed inline in a single paragraph using colon separation: Descriptive-Term (Original EponymTM, est. Year): Next-Term (EponymTM, est. Year).
- **Derivative Subdomain Rule:** If the number of distinct topical descriptors following a subdomain exceeds the available line length, the subdomain must be split into two or more derivative subdomains. Each new subdomain inherits the base label but is prepended with a substantively descriptive adjective that clarifies its conceptual focus. For example: [ALGORITHMS] may split into [HEURISTIC ALGORITHMS] and [DETERMINISTIC ALGORITHMS].
- **Domain Grouping:** Entries are grouped by their broad domain. Each domain is labeled with single quotes (e.g., 'MATHEMATICS' Approved Library). Within each domain section, related terms across its subdomains are listed using the compact library format.
- **No Redundancy:** Thematic keywords and term entries avoid redundancy. The keyword list provides conceptual orientation while the term entries provide specific replacements for eponyms.
- **Acronym Expansion:** All acronyms used as eponyms are expanded within the original name in parentheses. For instance, an entry for \"RSA\" will present the expanded form \"Rivest–Shamir–Adleman\" in the parentheses of the original name. This rule ensures that the historical names are fully transparent.
- **Consistent Syntax:** The grammar of each entry is constructed to provide clear attribution. Each term entry shows the descriptive replacement followed by the historical eponym and

establishment date in a consistent format.

- **Formatting:** Entries use minimal punctuation. Colons separate multiple entries within a subdomain's paragraph. The trademark symbol ™ is attached immediately after the historical name without a space, and the "est." date is formatted as a four-digit year (or a c. for circa if approximate).
- **Verification:** Each descriptive term is verified to ensure it does not introduce ambiguity and that it can replace the eponym in typical usage. The glossary generation process includes peer review to catch any descriptive names that might overlap with other concepts or lack clarity.

By following these structural rules, SDED v1.2's glossary achieves a high density of information while maintaining readability and consistency, thus making the Approved Vocabulary easy to adopt and reference across different disciplines.

4. Subdomain Expansion Rule

The **Subdomain Expansion Rule** governs how new subject areas are incorporated into the SDED vocabulary structure. As knowledge domains evolve, SDED must remain adaptable. The rule stipulates that subdomains can be added to a domain's bracketed list (or new domains introduced) when the volume or significance of new eponymous terms in that area warrants inclusion. The criteria are: - **Threshold of Inclusion:** A new subdomain should be introduced once there are a significant number of terms (e.g., five or more) from that area identified for approved naming, or if the subdomain represents an emerging field of high relevance (even with fewer terms). - **Maintaining Balance:** Any expansion should maintain the overall proportional coverage. When adding many terms in a new subdomain, consider adding or highlighting terms in other domains if needed to preserve a balanced cross-domain representation. - **Append-only Growth:** The structure of existing domains and subdomains is preserved for backward compatibility. New subdomains are appended to the domain's list, and new domain sections are appended to the document in alphabetical order (or logical order if a grouping is preferable). Existing entries are not shuffled between subdomains in a way that would change references from previous versions. - **Cross-Domain Entries:** If a term could belong to multiple domains or subdomains, it is placed under the domain where its usage is most dominant. Cross-references may be noted in future revisions, but duplication of entries is avoided to keep the vocabulary set streamlined. - **Versioning:** Any expansion of subdomains (or domains) triggers a version update. The version number (as in v1.2) reflects substantive additions. Minor additions or corrections within existing subdomains can be handled in patch versions (e.g., v1.2.1) but adding an entirely new subdomain would justify an increment to v1.3 or higher. - **Documentation:** New subdomains and their rationale for inclusion are to be documented in the Scope section of that version's specification, preserving a record of how the coverage of SDED evolves over time.

By enforcing the Subdomain Expansion Rule, SDED ensures that its growth is methodical and justified, preventing ad-hoc additions that could skew the focus or usability of the Approved Vocabulary. This rule helps maintain the integrity of the SDED structure as it expands, keeping it systematic and user-friendly.

Approved Libraries

Below are the Approved Libraries, organized by domain. Each library contains thematically grouped terms with their descriptive replacements and original eponymous names.

'MATHEMATICS' Approved Library

- **[ANALYSIS]:** Functions. Series. Continuity. Transformations. Spectra.
Frequency-Decomposition Transform (Fourier Transform™, est. 1822): Complex-Mapping Framework (Riemann Geometry™, est. 1851): Integral-Summation Series (Bernoulli Series™, est. 1713): Topological-Completeness Space (Hilbert Space™, est. 1902): Vector-Norm System (Banach Space™, est. 1922).

- **[ALGEBRA]:** Structures. Relations. Symmetry. Groups.

Polynomial-Permutation Structure (Galois Theory™, est. 1831): Invariant-Transformation Group (Lie Group™, est. 1873): Field-Construction Framework (Noether Theorem™, est. 1921): Curvature-Defined Smooth Geometry (Riemannian Geometry™, est. 1854).

- **[ARITHMETIC]:** Constants. Sequences. Patterns.

Recursive-Ratio Sequence (Fibonacci Sequence™, est. 1202): Prime-Distribution Function (Gaussian Integer Distribution™, est. 1801).

'PHYSICS' Approved Library

- **[MECHANICS]:** Motion. Force. Energy. Momentum.

Inertial Field Mechanics (Newtonian Mechanics™, est. 1687): Energy-Conservation Principle (Lagrangian Mechanics™, est. 1788): Wave-Interference Equation (Bernoulli Principle™, est. 1738).

- **[ELECTROMAGNETISM]:** Charge. Field. Potential. Induction.

Electromagnetic Field Equations (Maxwell Equations™, est. 1865): Magnetic-Flux Density Unit (Tesla™, est. 1880): Electric-Current Unit (Ampere™, est. 1820): Electric-Potential Unit (Volt™, est. 1860).

- **[THERMODYNAMICS]:** Temperature. Entropy. Equilibrium.

Thermal-Energy Scale Unit (Kelvin™, est. 1848): Entropy-Change Law (Carnot Cycle™, est. 1824).

'COMPUTER SCIENCE' Approved Library

- **[ALGORITHMS]:** Optimization. Search. Sorting. Graph Traversal.

Shortest-Path Optimization Algorithm (Dijkstra's Algorithm™, est. 1956): Divide-Merge Sorting Algorithm (Merge Sort™, est. 1945): Greedy Path Selection Algorithm (A*™, est. 1968).

- **[INFORMATION THEORY]:** Entropy. Coding. Noise. Redundancy.

Information-Entropy Theory (Shannon Information Theory™, est. 1948): Error-Correction Coding Scheme (Hamming Code™, est. 1950): Compression-Efficiency Model (Huffman Coding™, est. 1952).

- **[CRYPTOGRAPHY]:** Keys. Encryption. Authentication.

Public-Key Encryption Protocol (RSA™, est. 1977): Symmetric-Block Cipher Standard (AES™, est. 1998): Hash-Generation Function (SHA™, est. 1993).

'ENGINEERING' Approved Library

- **[ELECTRICAL]:** Signals. Circuits. Frequency. Amplification.

Frequency-Response Analysis (Bode Plot™, est. 1938): Sampling-Rate Criterion (Nyquist Theorem™, est. 1928): Oscillation-Frequency Standard (Hertz™, est. 1888).

- **[STRUCTURAL]:** Stress. Strain. Elasticity. Load.

Elastic-Deformation Law (Hooke's Law™, est. 1678): Structural-Failure Criterion (Mohr-Coulomb Theory™, est. 1773).

'MEDICINE' Approved Library

- **[NEUROLOGY]:** Cognition. Degeneration. Motor Control.

Neural-Degenerative Condition (Alzheimer's Disease™, est. 1906): Motor-System Disorder (Parkinson's Disease™, est. 1817).

- **[IMMUNOLOGY]:** Immune Response. Inflammation. Cell Activity.

Lymphatic-Proliferation Condition (Hodgkin Lymphoma™, est. 1832): Autoimmune Inflammatory Condition (Crohn's Disease™, est. 1932).

- **[GENETICS]:** Chromosomal. Hereditary. Syndromes.

Trisomy-21 Condition (Down Syndrome™, est. 1866): Cystic-Fibrosis Condition (Cystic Fibrosis™, est. 1938).

'LINGUISTICS AND HUMANITIES' Approved Library

- **[PHILOSOPHY OF LANGUAGE]:** Structure. Meaning. Syntax. Semantics.

Generative-Syntax Framework (Chomskyan Linguistics™, est. 1957): Structural-Semantics Model (Saussurean Linguistics™, est. 1916).

- **[ECONOMICS]:** Markets. Production. Policy.

Macroeconomic-Regulation Framework (Keynesian Economics™, est. 1936): Value-Labor Analysis (Marxist Economics™, est. 1867).